C24 Reactions of Ions in Aqueous Solution – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Aqua Ions and Hydroxides **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Colour of $[Fe(H_2O)_6]^{2+}$				Pale green
2	Colour of $[Fe(H_2O)_6]^{3+}$				Yellow-brown
3	Colour of $[Cu(H_2O)_6]^{2+}$				Blue
4	Colour of $[Al(H_2O)_6]^{3+}$				Colourless
5	Why M^{3+} aqua ions are more acidic than M^{2+}				Higher charge density polarises water
6	Precipitate of Fe^{2+} with NaOH				Green $Fe(H_2O)_4(OH)_2$
7	Precipitate of Fe^{3+} with NaOH				Brown $Fe(H_2O)_3(OH)_3$
8	Precipitate of Cu^{2+} with NaOH				Blue $Cu(H_2O)_4(OH)_2$
9	Precipitate of Al^{3+} with NaOH				White $Al(H_2O)_3(OH)_3$
10	$Al(OH)_3$ with excess NaOH				Dissolves: colourless $[Al(OH)_4]^-$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Ammonia, Carbonate and Chloride **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Cu^{2+} with excess NH_3				Deep blue solution $[Cu(NH_3)_4(H_2O)_2]^{2+}$
2	Fe^{2+} with NH_3				Green precipitate
3	Fe^{3+} with NH_3				Brown precipitate
4	Al^{3+} with excess NH_3				White precipitate, insoluble in excess
5	Cu^{2+} with Na_2CO_3				Blue-green precipitate $CuCO_3$
6	Fe^{2+} with Na_2CO_3				Green precipitate $FeCO_3$
7	Fe^{3+} with Na_2CO_3				Brown precipitate and effervescence
8	Al^{3+} with Na_2CO_3				White precipitate and effervescence
9	Why M^{3+} with carbonate effervesces				Acidic solution releases CO_2
10	Cu^{2+} with conc HCl				Yellow-green solution $[CuCl_4]^{2-}$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 3: Equations **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Hydrolysis of $[Fe(H_2O)_6]^{3+}$				$[Fe(H_2O)_6]^{3+} + H_2O \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+} + H_3O^+$
2	$[Cu(H_2O)_6]^{2+}$ with OH^-				$[Cu(H_2O)_6]^{2+} + 2OH^- \rightarrow Cu(H_2O)_4(OH)_2 + 2H_2O$
3	$[Fe(H_2O)_6]^{3+}$ with OH^-				$[Fe(H_2O)_6]^{3+} + 3OH^- \rightarrow Fe(H_2O)_3(OH)_3 + 3H_2O$
4	$Al(H_2O)_3(OH)_3$ with excess OH^-				$Al(H_2O)_3(OH)_3 + OH^- \rightarrow [Al(OH)_4]^- + 3H_2O$
5	$Cu(OH)_2$ with excess NH_3				$Cu(H_2O)_4(OH)_2 + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2OH^- + 2H_2O$
6	$[Cu(H_2O)_6]^{2+}$ with conc HCl				$[Cu(H_2O)_6]^{2+} + 4Cl^- \rightarrow [CuCl_4]^{2-} + 6H_2O$
7	$[Cu(H_2O)_6]^{2+}$ with CO_3^{2-}				$[Cu(H_2O)_6]^{2+} + CO_3^{2-} \rightarrow CuCO_3 + 6H_2O$
8	$[Fe(H_2O)_6]^{3+}$ with CO_3^{2-}				$2[Fe(H_2O)_6]^{3+} + 3CO_3^{2-} \rightarrow 2Fe(H_2O)_3(OH)_3 + 3CO_2 + 3H_2O$
9	Amphoteric hydroxide				$Al(OH)_3$
10	Lewis acid				Electron pair acceptor

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.